

The Overfitting Theorem

Expected maximum Sharpe, minimum backtest length, compensation effects

After Bailey, Borwein, López de Prado & Zhu, Notices of the AMS 61(5), May 2014, 458–471 · DOI 10.1090/noti1105

SOURCE

The published article is open access and free at <https://www.ams.org/notices/201405/rnoti-p458.pdf> — note the hyphen in rnoti-p458. Read it. These four pages are my own reconstruction of its mathematical spine, written to be checkable line by line and to connect each result to a gate in the build. No text from the paper is reproduced here.

SETUP

1. The question

A researcher holds T periods of returns and evaluates N configurations of a strategy. She keeps the best. The question the theory answers is not whether the winner is profitable. It is:

How large an in-sample Sharpe ratio should we expect from the best of N configurations when every one of them has true Sharpe zero?

If that quantity is large, then an impressive backtest carries no information, and the entire practice of selecting on in-sample performance collapses. It is large.

Assumptions

Returns are i.i.d. normal with mean μ and variance σ^2 . The annualised Sharpe is $SR = (\mu/\sigma)\sqrt{q}$ with q periods per year. Since μ and σ are unknown, we work with the estimate built from sample moments. Lo (2002) gives its asymptotic distribution: for y years of data,

$$SR \rightarrow N(SR, (1 + SR^2/(2q)) / y)$$

Set $\mu = 0$ and $y = 1$. Then $SR = 0$, and critically the variance becomes 1 regardless of q . Sampling more frequently does not help. Each configuration's estimated annualised Sharpe is a draw from a standard normal, and the researcher's chosen result is the maximum of N such draws.

RESULT ONE

2. The expected maximum

PROPOSITION 1

For $z_1, \dots, z_N$ i.i.d. standard normal and N large,

$$E[\max z] \approx (1-\gamma) \cdot \Phi^{-1}(1 - 1/N) + \gamma \cdot \Phi^{-1}(1 - 1/(N \cdot e))$$

with $\gamma \approx 0.5772156649$ the Euler-Mascheroni constant, and the upper bound $E[\max z] < \sqrt{2 \ln N}$.

Why it is true

The argument is pure extreme-value theory and takes three steps.

- Domain of attraction. By Fisher-Tippett-Gnedenko, a normalised sample maximum converges to one of three limit laws. The normal distribution lies in the Gumbel maximum domain of attraction, so $(\max z - \alpha)/\beta$ converges to the standard Gumbel law with CDF $\exp(-\exp(-x))$.
- Normalising constants. The standard choice is $\alpha = \Phi^{-1}(1 - 1/N)$ and $\beta = \Phi^{-1}(1 - 1/(N \cdot e)) - \alpha$. Read α as the $1-1/N$ quantile — the level the maximum of N draws typically reaches — and β as the width of the fluctuation around it.
- Take the expectation. The mean of the standard Gumbel is exactly γ . So $E[\max z] \approx \alpha + \gamma\beta$, which expands to the expression above.

The bound $\sqrt{2 \ln N}$ is the more memorable form and comes from the standard tail estimate on the normal maximum. Carry it: the best of N noise results grows like the square root of the logarithm of N . Slowly, but without bound, and fastest in the regime a solo researcher occupies — the move from $N = 1$ to $N = 100$.

Numerically

N	$E[\max z]$	$\sqrt{2 \ln N}$	MinBTL (yrs) at $SR^* = 1$	Best annualised SR from noise, $T = 1008$
5	1.193	1.794	1.4	0.60
10	1.575	2.146	2.5	0.79
45	2.236	2.759	5.0	1.12
100	2.531	3.035	6.4	1.27
500	3.053	3.526	9.3	1.53
1,000	3.255	3.717	10.6	1.63
8,800	3.829	4.262	14.7	1.91

Computed independently via `scipy.stats.norm.ppf`. Verification: the source paper states that $N = 10$ trials yields an expected in-sample Sharpe of 1.57; this implementation returns 1.575. It also states that five years of data supports at most about forty-five independent configurations; row three gives MinBTL = 5.0 years at $N = 45$. Both reproduce, so the implementation is correct.

RESULT TWO

3. Minimum backtest length

Proposition 1 assumed one year. For y years, the standard error of the annualised Sharpe scales as $y^{-1/2}$, so the expected best-of- N in-sample Sharpe becomes

$$E[\max SR] \approx y^{-1/2} \cdot [(1-\gamma)\Phi^{-1}(1-1/N) + \gamma\Phi^{-1}(1-1/(Ne))]$$

Solve for y .

THEOREM 2 — MinBTL

The minimum backtest length, in years, needed to avoid selecting a strategy with in-sample Sharpe $E[\max SR]$ from N independent configurations whose true out-of-sample Sharpe is zero:

$$\text{MinBTL} \approx [(1-\gamma)\Phi^{-1}(1-1/N) + \gamma\Phi^{-1}(1-1/(Ne))]^2 / E[\max SR]^2 < 2 \ln(N) / E[\max SR]^2$$

The quadratic denominator is what makes this bite. Halving your Sharpe target quadruples the history required. And the consequences at ordinary scales are severe: with five years of daily data you may try roughly forty-five independent configurations before you are all but guaranteed to manufacture an in-sample Sharpe of 1 with a true out-of-sample Sharpe of 0. With two years, about seven.

Seven. A researcher who tries a 3×3 parameter grid on two years of data has already exhausted her statistical budget.

Model complexity

N grows combinatorially in the number of parameters. A rule with k binary switches gives $N = 2^k$. Seven switches is 128 configurations, and by the table above that supports an expected in-sample Sharpe above 2.5 on pure noise. No supercomputer is required and no sophistication is required — the paper's point is precisely that perfect fitting is computationally impossible while sufficient fitting is trivial. You do not need to search 2^{100} configurations. You need to search about a hundred.

Which yields the sentence the whole paper turns on: a backtest that does not report N cannot be assessed for overfitting risk, even in principle. The information required for the inference is absent. This is why the trials ledger exists in this build and why N is written by the harness rather than typed by you.

RESULT THREE

4. Compensation effects, and a correction

Here is where I stated things too strongly in the earlier playbooks, so take this as the corrected version. Overfitting does not always produce negative out-of-sample performance. It depends on whether the process has memory.

Case A — no memory

Simulate N driftless Gaussian random walks, split each into in-sample and out-of-sample halves, and select the configuration with the highest in-sample Sharpe. The selected path's in-sample Sharpe lands somewhere around 1.2 to 2.6. Its out-of-sample Sharpe is distributed around zero. Selection bought you nothing, and it cost you nothing. The scatter of in-sample against out-of-sample Sharpe is a structureless circular cloud centred at the origin.

Case B — memory, and the anticorrelation result

Now impose a compensation effect. Two ways to create one, both realistic. Either recentre each path to a common mean, removing one degree of freedom, or make the process serially correlated — an AR(1) with coefficient ϕ . Real financial series have both: overcrowded trades unwind, bubbles burst, flows reverse, cycles turn.

PROPOSITIONS 3 AND 5

For two configurations A and B of the same model with equal standard deviations across both sub-samples, under either a global mean constraint or a stationary AR(1) process:

$$SR_{IS}(A) > SR_{IS}(B) \implies SR_{OOS}(A) < SR_{OOS}(B)$$

An exact reversal, not a weakening. Whichever configuration looks better in-sample is guaranteed to look worse out-of-sample.

The mechanism is conservation. Under a global constraint the sample mean is fixed, so an above-average first half forces a below-average second half; the algebra is two lines. Under AR(1) the same logic runs through the level: a high in-sample Sharpe means the process ended the in-sample window far above equilibrium, and mean reversion then drags it back during the out-of-sample window. The relevant timescale is the half-life,

$$\tau = -\ln(2) / \ln(\phi)$$

which for $\phi = 0.995$ is about 138 observations — comfortably inside a typical out-of-sample window, which is exactly why the effect shows up in practice.

THE CORRECTED CLAIM

Overfitting yields zero expected out-of-sample performance in a memoryless process and negative expected out-of-sample performance whenever compensation effects are present. Financial series have compensation effects. So the customary disclaimer that past performance does not indicate future results is, for an overfit backtest, too generous: strong in-sample performance is a positive indicator of negative future results.

CONSEQUENCE

5. What this obliges the build to do

Four obligations follow directly, and each is already a gate.

Result	Obligation	Gate
E[max] grows like $\sqrt{(2 \ln N)}$	The null is not zero. Calibrate it empirically with turnover-matched random strategies and require the real result to beat that distribution.	G6
MinBTL	Compute it from the ledger's N before publishing any Sharpe. If $T < \text{MinBTL}$, the headline reads uninterpretable, not impressive.	G9
Anticorrelation under memory	Plot in-sample against out-of-sample Sharpe across the whole grid. A negative slope means you have reproduced Proposition 5 on your own data — report it.	G8
N must be disclosed	Machine-written trials ledger, append-only. Estimate effective N by dimension reduction on the trial correlation matrix, since the propositions assume independence and yours are not independent.	—

One last number to hold onto. The paper's own worked example searches a four-parameter monthly trading rule — entry day, holding period, stop loss, side — across 8,800 combinations, on a series of 1,000 daily prices that is a pure random walk by construction. The winning configuration posts an annualised Sharpe of 1.27, and its Probabilistic Sharpe Ratio statistic comes out at 2.83, implying under one percent probability that the true Sharpe is below zero. Every part of that is a lie produced by honest arithmetic. Nothing was there.

PROTOCOL

6. Reproduce all three results yourself

Do this before writing engine code. It needs numpy and scipy and nothing else, it runs in under a minute, and it is the cheapest way to convert these propositions from things you have read into things you have seen.

Experiment A — Proposition 1

Draw $N = 1000$ standard normals, take the maximum, repeat 10,000 times, average. Compare against the closed form. Expect agreement to roughly two decimal places, with the empirical mean near 3.24 against a predicted 3.255. Then sweep N across four orders of magnitude and confirm the growth tracks $\sqrt{(2 \ln N)}$ from below.

Experiment B — the memoryless case

Generate $N = 1000$ driftless Gaussian random walks of length $T = 1000$. Split each at the midpoint. Scatter in-sample Sharpe against out-of-sample Sharpe. Expect a structureless circular cloud centred on the origin, and a regression slope statistically indistinguishable from zero. Now select the single path with the highest in-sample Sharpe and look at its out-of-sample half. Expect in-sample somewhere in 1.2 to 2.6, out-of-sample scattered around 0.

Experiment C — compensation effects

Rerun Experiment B twice more. First recentre every path to a common mean before splitting. Second, replace the random walk with a stationary AR(1) at $\phi = 0.995$. Expect in both cases a visibly negative slope in the in-sample versus out-of-sample scatter, with the slope coefficient significant at any conventional level while the intercept is not. That negative slope is Propositions 3 and 5 rendered as a picture, and it is the single most important plot in this entire subject.

THE ONE THING TO CARRY AWAY

Experiment B says a lucky backtest costs you nothing. Experiment C says that in a market with memory — which is every market — the same lucky backtest costs you money, in expectation, in proportion to how hard you searched for it. The harder you optimised, the worse you will do. That is the result, and no amount of out-of-sample testing rescues you from it if you never counted N .

Original exposition after Bailey, Borwein, López de Prado and Zhu (2014). Propositions are restated and re-derived in my own words; no text is reproduced. Numerical tables computed independently and cross-checked against the values stated in the source. Read the original at the AMS link on page 1.